

Face variation: Reputation-based Reward System & Manual Validation Framework

Phase 4, Cycle c2(end) – May 2, 2025

Post-Phase 4 — Cycle 2 Updates (April 20, 2026)

Status of Manual Grading

As of April 20, 2026, the Face Variations collection phase for Phase 4 — Cycle 2 has officially concluded. Since that date, the team has shifted full focus to the manual grading pipeline for Faces and to the integration of screen replays into the validation workflow.

At this point, all submitted Face Variations have been processed through the grading workflow. The same evaluation criteria from the previous cycle continue to apply — the steps remain unchanged for consistency and fairness:

Step 1: Filter out all corrupt and duplicate images.

Step 2: Confirm identity is relatively preserved (AdaFace).

Step 3: Run the image through variation-type detection.

Step 4: Manual validator checks for:

- Is the person still the same individual?
- Did the miner follow the query?
- Did the miner exploit?
- Does the image have more than one variation?
- Are artifacts, overlays, or shortcuts present?

Screen replays were added as a new validation signal in this cycle. Replays let the team verify miner workflows end-to-end during manual review, and the results have been very positive — they made it noticeably easier to confirm that variations were genuinely produced rather than copy-pasted, looped, or otherwise shortcut.

Note: This cycle does not include UAVs. Final scoring is therefore based entirely on the Face Variation evaluation.

Time-Based Rewards and Reputation

We continue to reward miners who have consistently contributed high-quality, non-exploitative data over multiple phases. The bonus structure for Phase 4 — Cycle 2 is:

Participation history	Age Factor	Bonus
Phase 3 — Cycle 1 **and** Phase 4 — Cycle 1 (and active now)	1.20	+20%
Phase 4 — Cycle 1 only (and active now)	1.10	+10%

The bonus is incorporated through the Age Factor in the score formula below.

Final Score Calculation

Because there is no UAV component this cycle, the formula simplifies to:

Total Score = (Remaining Rewards from Phase 4 – Cycle 1) + (Face Variation Score × Age Factor)

Where:

- Remaining Rewards from Previous Cycle carries forward any unrealized rep_score from Phase 4 — Cycle 1.
- Face Variation Score is computed from the Δ rep mapping table (unchanged, see below).

- Age Factor is `1.20`, `1.10`, or `1.00` based on the participation history described above.

Slight Change in the Rewards Mechanism

We made two adjustments to the reward mechanism for this cycle: a new Platinum tier and a faster decay rate. Full technical detail lives in the UAV rewards report; the summary is below.

New Platinum tier (extended top-end normalization)

Tier	Multiplier
 Platinum	1.25
 Diamond	1.15
 Gold	1.10
 Silver	1.05
 Bronze	1.02
 Neutral	1.00
!! Watch	0.90

The normalization curve was also extended at the top so that high-rep miners are no longer compressed near the cap. Platinum miners now have explicit headroom (raw 200+ → normalized 2.00–3.00).

Faster decay

Decay per query: 0.01 → 0.02

Why we changed this

Over the last two cycles, some miners did not receive their full Face variation rewards within the cycle window. We carried the remainder forward by adding it to their score in the following cycle. In practice we found this unfair: rolling unpaid rewards into future scores meant some miners might never fully clear what they were owed in a single cycle, and the accounting felt perpetual rather than settled.

Increasing rep-score decay shortens the reward horizon and aligns it with how often miners are evaluated. That is fairer to miners who actually earned rewards in the current window — emissions are realized on a clearer timetable instead of being indefinitely deferred through score carryover. It also prevents the leaderboard from freezing in place: top miners cannot stay dominant forever purely from slow score drainage, and after a cycle ends, everyone starts from a more level playing field for the next one.

Leaderboard Growth

Phase 4 - Cycle 2 saw a substantial increase in miner participation. The leaderboard is significantly larger and more diverse than the previous cycle:

Metric	Phase 4 - Cycle 1	Phase 4 - Cycle 2
Miners on the leaderboard	8	100

Tier breakdown for Phase 4 - Cycle 2:

Tier	Count
Platinum	1
Diamond	4
Gold	2
Silver	1
Bronze	7
Neutral	78
Watch	7

Because of the size of the leaderboard, only the ****top 15**** miners are listed below

Miner's Reputation Leaderboard (Top 15)

Reputation is determined by validated Face performance. Miners not listed here either fall into the long Neutral tail, the Watchlist, or did not score high enough to surface.

Rank	Hotkey	Score	Tier
1	5Hmz	285.95	Platinum
2	5DUy	161.89	Diamond
3	5GsC	140.88	Diamond
4	5FqF	108.61	Diamond
5	5C5h	63.71	Diamond
6	5Cev	45.67	Gold
7	5EBx	35.49	Gold
8	5GP1	28.88	Silver
9	5FCD	11.62	Bronze
10	5CS3	11.18	Bronze
11	5Dks	8.50	Bronze
12	5Fzs	8.43	Bronze
13	5GQr	8.19	Bronze
14	5GZd	6.30	Bronze
15	5EPi	5.02	Bronze

Δrep Mapping Table (unchanged)

Validation score	Meaning	Δrep
+5	Perfect	+0.1
+4	Near Perfect	+0.04
+3	Good	+0.03

+2	Okay	+0.02
+1	Decent	+0.01
0	Unclear	0.00
-1	partial spam	-0.05
-2	Misleading	-0.08
-3	Duplicate	-0.10
-4	Not enough info	-0.25
-5	Cheat	-0.5

Top 2 Miner's Score Break

The following table shows how raw UAV outcomes translated into reputation for the top two miners. Each category multiplies by weight depending on the severity or quality of the submission.

Hotkey: 5Hmz...BE63

Score	-5	-4	-3	-2	-1	0	1	2	3	4	5	total
Count	0	0	1	3	2	13	76	355	1177	2509	498	4634
Δrep	-0.5	-0.25	-0.1	-0.08	-0.05	0	0.01	0.02	0.03	0.04	0.1	
Total	0	0	-0.1	-0.24	-0.1	0	0.76	7.1	35.31	100.36	49.8	192.89

Face score = 192.89

Last cycle score = 54.48

Age = 3 = 1.2 (age factor)

Age score = $192.89 * 1.2 = 231.47$

Rep score = last cycle score + age score = 285.95

Hotkey: 5GsC... Kybn

Score	-5	-4	-3	-2	-1	0	1	2	3	4	5	total
Count	0	0	0	3	9	200	484	739	1024	1614	207	4280
Δrep	-0.5	-0.25	-0.1	-0.08	-0.05	0	0.01	0.02	0.03	0.04	0.1	
Total	0	0	0	-0.24	-0.45	0	4.84	14.78	30.72	64.56	20.7	134.91

Face score = 134.91

Last cycle score = 0

Age = 3 = 1.2 (age factor)

Age score = 134.91 * 1.2 = 161.89

Rep score = last cycle score + age score = 161.89

Score Normalization Ranges

Each tier has a bounded window mapping raw reputation into a reward-safe range.

Tier	Raw Range	Normalized Range
Watch	0.10–0.699	0.50–0.70
Neutral	0.70–1.00	0.70–1.00
Bronze	1.00–1.999	1.00–1.20
Silver	2.00–9.999	1.20–1.50
Gold	10–49.99	1.50–1.80
Diamond	50–200	1.80–2.00
Platinum	200-999	2.00-3.00

Weight Shift: KAV vs Face Variation/UAV

We also updated the score weighting to put more emphasis on adversarial image performance:

- KAV weight: 0.15 → 0.10
- Face Variation/UAV weight: 0.85 → 0.90

Previous formula:

$$\text{Miner_rep} = 0.85 \cdot \Delta\text{face_score} + 0.15 \cdot \Delta\text{rep_KAV}$$

New formula:

$$\text{Miner_rep} = 0.90 \cdot \Delta\text{face_score} + 0.10 \cdot \Delta\text{rep_KAV}$$

This change reflects Phase 4 priorities: stronger focus on Face Variation and UAV signal, and less emphasis on baseline KAV contribution.

Score Sheet

Score	Condition / path	Status
-5	Duplicate-level (copy and paste images into different backgrounds instead of synthetic generation)	rejected
-4	Severe aspect-ratio distortion (variation face >15.5% narrower than seed; delta < -0.155)	rejected
-3	Face identity below threshold or duplicate with other images (based on hash); or MediaPipe face count is not exactly 1 (0 or >1); religious head covering + gender-specific request but seed gender mismatch	recommended_scored / rejected
-2	No variation detected (claimed variation type but comparison detected no variation); or duplicate with other images (duplicate similarity checker)	rejected
-1	Duplicate with seed image (hash match); or variation resolution $\leq 512 \times 672$; or face not dominant part of image; or background accessory path with insufficient match and no background detected (0-1 = -1)	recommended_scored / rejected
0	Download/write failed; comparison failed (manual review); label mismatch with variation detected (rejected); background accessory path insufficient match	rejected / manual_review / recommended_scored
2	Background accessory path: would be 3 but no background change detected (3-1)	recommended_scored
3	Standard: identity 0.6-0.95 with partial variation match; background accessory: conf ≥ 0.5 (matching or other accessory)	recommended_scored
4	Background accessory: would be 5 but no background detected (5-1)	recommended_scored
5	Standard: identity ≥ 0.6 with full variation match; background accessory: matching accessory + ada_sim ≥ 0.6 + conf ≥ 0.5	recommended_scored

Screen replay score sheet:

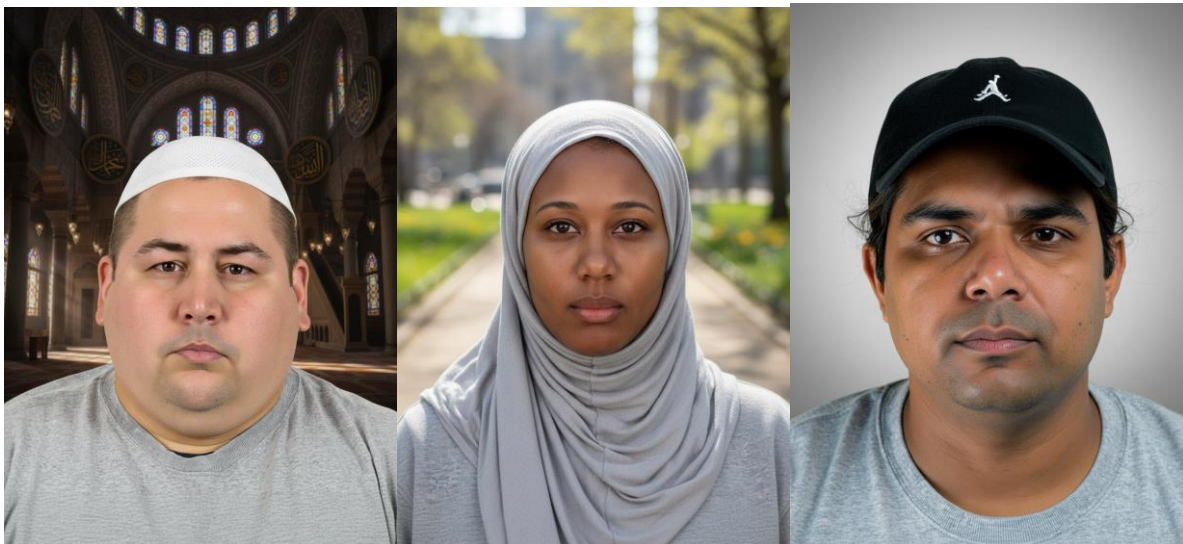
Score	Condition / path	Status
Info	The 3 checks are: cue 1 present, cue 2 present, and device present.	reference
-3	<i>screen_replay</i> : face identity below threshold or duplicate with other images (based on hash); or MediaPipe face count is not exactly 1 (0 or >1);	recommended_scored / rejected
-2	<i>screen_replay</i> : duplicate with other images (duplicate similarity checker)	rejected
-1	<i>screen_replay</i> : duplicate with seed image (hash match); or variation resolution $\leq 512 \times 672$	recommended_scored / rejected
0	<i>screen_replay</i> : download/write failed; comparison failed (manual review); label mismatch with variation detected (rejected); identity < 0.4 (hard gate); all 3 checks missing; or detection failed (manual review)	rejected / manual_review / recommended_scored
1	<i>screen_replay</i> : identity ≥ 0.6 with only 1 of the 3 checks present (2 missing)	recommended_scored
3	<i>screen_replay</i> : identity ≥ 0.6 with 2 of the 3 checks present (1 missing)	recommended_scored
5	<i>screen_replay</i> : identity ≥ 0.6 and all 3 checks pass (cue 1 + cue 2 + device)	recommended_scored

Examples

Screen reply



Background



Pose



Lighting



Expression



Negative examples (common pit-falls)



(Face is not dominant)



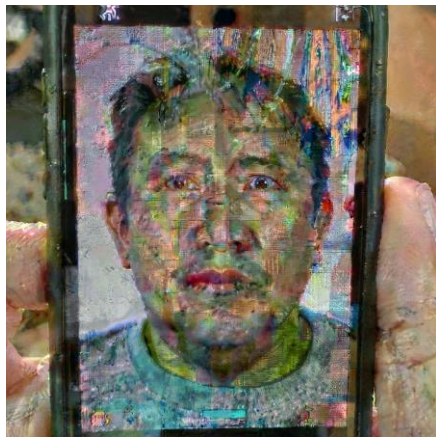
(No accessories)



(Have more than one accessory and gender does not align)



(Gender miss match (women do not typical wear turbans))



(color disfiguration to the point where face not detect)

Post-Phase4-Cycle 1 – Updates (March 2, 2026)

Status of Manual Grading

As of March 2, 2026, the Face Variations collection phase has officially concluded. Since that date, the team has shifted its full focus to completing the manual grading pipeline for all submitted Face Variations.

At this point, all submitted Face Variations have been processed through the grading workflow. The same evaluation criteria previously outlined in earlier updates—referenced in other sections of this document—continue to apply throughout the grading process. These criteria ensure consistency, fairness, and alignment with the project's overall validation standards.

In earlier updates, we provided an overview of how the manual validation pipeline operates. This section expands on that explanation by providing additional details on scoring progress and by introducing the reputation tiers that will be used to determine emissions.

Within this document, individual Face scores will be released, along with the combined total score calculated from both the Face evaluation and the associated UAV address score. This combined score determines the final reputation adjustment.

For reference, the calculation formula used to determine the final reputation change is as follows:

$$\Delta\text{rep}_{p4} = 0.95 \cdot \Delta\text{rep}_{\text{face}} + 0.05 \cdot \Delta\text{rep}_{\text{UAV_address}}$$

This formula reflects the weighting of the two components:

- 95% of the final reputation score is derived from the Face evaluation.
- 5% of the final reputation score is derived from the UAV address reputation.

This weighted approach ensures that the quality and validity of the submitted Face Variations remain the primary factor in scoring, while still incorporating a smaller contribution from the associated address reputation.

Time-Based Rewards and Reputation

In this cycle, we are introducing a new feature designed to reward miners who have consistently contributed high-quality, non-exploitative data over an extended period of time. This feature recognizes long-term participation and sustained contributions to the network.

Any miner who participated in Phase 3 – Cycle 1 and Phase 4 – Cycle 1 will receive a 10% bonus applied to their score. This adjustment reflects the value of continued participation and reliable data contributions across multiple phases of the project.

The bonus is incorporated into the overall scoring system through an age factor, which increases the final score for qualifying miners.

Final Score Calculation

The total score used to determine rewards and reputation adjustments is calculated using the following formula:

Total Score = (Remaining Rewards from Phase 3 – Cycle 1) + ((UAV Address Rewards × 0.05) + (Face Variation Score × 0.95)) × Age Factor

This formula maintains the same weighting structure described earlier in the document:

- 95% of the score is derived from the Face Variation evaluation.

- 5% of the score is derived from the UAV address reputation.

The Age Factor is applied afterward to reward miners who participated in both Phase 3 – Cycle 1 and Phase 4 – Cycle 1, providing a 10% bonus to their final score.

Miner's Reputation Leaderboard

Reputation is determined by validated Face performance. Miners not listed here fall into the Normal, barely participated, or Watchlist, with negative or near-zero scores due to spam, duplicates, invalid coordinates, or consistently low-value UAVs.

Rank	hotkey	score	tier
1	5HmzMBNbEgjTxzxzbpg42hkrM5gzvPcaApD1kKP5nEq8BE63	115.97	 Diamond
2	5GsChcZ5FBfdbJRX4cTRgke5upQ3tQ24Z1K3r7PgojD9Kybn	28.87	 Silver
3	5DUyTmjJ5P2xFHg6qBJW4FkbjKrwLwg7SvU8UcwdqYiEU39a	8.36	 Bronze
4	5CS3nxrpwCkrbv3PYQD6LLDtxEickHq21sJ9FPAvewkYpouY	5.59	 Bronze
5	5GZdCsRkiXJ5Kcks7iXZhAgC4aLna3r5JUhcJkHkL1RTPr51	2.94	 Neutral
6	5C7MdoMGTpgRBNEodu6XkNWJn7zJHAmAKnz4VjLJUL8JmT3D	2.35	 Neutral

7	5G4jmgfWjapk4amvVCciebCqav4yPjtR3Jp3ZJUJ3d6beeKG	2.38	 Neutral
8	5DFmJzd32EqKJFT9XUZFYDB6MPahQDP38WHACeToNGQ2NK5e	2.05	 Neutral

Top 2 Miner's Score Break

The following table shows how raw UAV outcomes translated into reputation for the top two miners. Each category multiplies by weight depending on the severity or quality of the submission.

Hotkey: 5Hmz...BE63

Score	-5	-4	-3	-2	-1	0	1	2	3	4	5	total
Count	18	0	3	175	37	499	634	577	1572	377	600	4442
Δrep	0.5	0.25	0.1	-0.08	-0.05	0	0.01	0.02	0.03	0.04	0.1	
Total	-9	0	-0.3	-14	-1.85	0	6.34	11.5	47.2	15	60	115.97

remaining rewards from phase 3 cycle 1 = 0

UAV address rewards = -171.2

Face variation = 115.97

age factor = 1.1

Total score: **111.77** = (0) + ((-171.2 * 0.05) + (115.97 * 0.95) * 1.1)

Hotkey: 5GsC... Kybn

Score	-5	-4	-3	-2	-1	0	1	2	3	4	5	total
Count	0	0	4	9	18	121	217	159	304	193	77	1102
Δrep	0.5	0.25	0.1	-0.08	-0.05	0	0.01	0.02	0.03	0.04	0.1	

Total	0	0	-0.4	-0.72	-0.9	0	2.17	3.18	9.12	7.72	7.7	28.87
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remaining rewards from phase 3 cycle 1 = 0

UAV address rewards = 0

Face variation = 28.87

age factor = 1

Total score: **27.43** = $(0) + ((0 * 0.05) + (28.87 * 0.95) * 1)$

The table below, Δ rep Mapping Table, shows how each validation score contributes to a miner's reputation, with positive scores increasing Δ rep and negative scores reducing it.

Validation score	Meaning	Δ rep
+5	Perfect	+0.1
+4	Near Perfect	+0.04
+3	Good	+0.03
+2	Okay	+0.02
+1	Decent	+0.01
0	Unclear	0.00
-1	partial spam	-0.05
-2	Misleading	-0.08

-3	Duplicate	-0.10
-4	Not enough info	-0.25
-5	Cheat	-0.5

Reputation Tiers & Normalization (How Miner Rewards Are Calculated)

Reputation is not used raw. Instead, miners are assigned a tier and their scores are normalized before being applied to emissions. This ensures fairness and prevents extremely high scores from dominating the incentive pool.

Tier Multipliers

These increase or decrease a miner’s effective reputation:

Tier	Multiplier
 Diamond	1.15
 Gold	1.10
 Silver	1.05
 Bronze	1.02
 Neutral	1.00
 Watch	0.90

Score Normalization Ranges

Each tier has a bounded window mapping raw reputation into a reward-safe range.

Tier	Raw Range	Normalized Range
Watch	0.10–0.699	0.50–0.70
Neutral	0.70–1.00	0.70–1.00

Bronze	1.00–1.999	1.00–1.20
Silver	2.00–9.999	1.20–1.50
Gold	10–49.99	1.50–1.80
Diamond	50–9999	1.80–2.00

The final Face/UAV reward is Normalized Score × Tier Multiplier, which is then will be the 80% of the final score, combined with KAV quality (20%), and the final score will be the score the validator will set weight on the chain based on.

Below is a **clean section written to match the tone and structure of the rest of your documentation**. It keeps the same explanatory style, avoids casual language, and clearly explains both **what was discovered** and **how duplicates were handled**.

Duplicate Submissions and Multi-Hotkey Activity

During the manual validation and post-processing stage of the Face Variations pipeline, the team also performed cross-miner duplication analysis. This process was designed to detect situations where the same underlying model or workflow was producing nearly identical outputs across multiple miners.

The goal of this analysis is to ensure that the network rewards **unique and independently generated data**, rather than scaled replication of the same output across multiple miners.

Real Example From Cycle 1

A clear example illustrates why this policy is necessary.

In Cycle 1, a single operator ran five separate hotkeys (5GsC, 5EWg, 5HZF, 5FyA, and 5ERu). All five performed well in the initial scoring results and each reached the Silver tier, with scores of 28.87, 28.57, 28.27, 26.10, and 25.53, placing 2nd through 6th on the leaderboard.

These scores placed the five hotkeys **2nd through 6th on the leaderboard**.

If these submissions had been considered independent and unique, the five hotkeys combined would have earned approximately **17,942 a over roughly 40 days** as their scores gradually decayed through validator reward allocation.

Duplication Detection Results

However, after running the duplication detection pipeline, we discovered that the five hotkeys were submitting **nearly identical images**.

In practice, this meant the same model was producing the same outputs and distributing them across multiple miners. Because the images were not uniquely generated for each submission, the system classified these results as **duplicate data**.

Duplicate data undermines the purpose of the validation pipeline, which is designed to reward **unique transformations and independent contributions**.

Penalty Outcome

As a result, all five hotkeys were penalized and placed on the **Watch tier**, causing their reputation scores to drop to near zero.

In the subsequent leaderboard snapshot:

- Only one hotkey briefly remained visible with a score of **0.15**
- That hotkey earned approximately **34 a for a single day**
- After that, all five hotkeys effectively disappeared from the leaderboard

Why Running Multiple Hotkeys Backfired

Ironically, the miner would have earned significantly more by operating a **single hotkey**.

With their highest original score of **28.87**, a single miner would have:

- earned **higher daily rewards**, since they would not be competing with their own duplicated outputs
- remained eligible for **40+ days** as reputation gradually decayed
- likely earned **approximately 8,500–10,000 α** in total rewards

Instead, by splitting the same model across five hotkeys and triggering duplication detection, the miner's expected outcome shifted from a realistic **~9,000 α** to **essentially 0 α**.

How Duplicate Submissions Were Handled

To handle these cases fairly, the system applied a deterministic duplicate resolution policy.

When identical or near-identical submissions were detected across multiple miners:

- **One hotkey was selected to retain the positive score that comes under the same cold key as the other hotkeys that submitted duplicate data.**
- **The duplicate submissions were assigned negative scores**

Importantly, the system **consistently selected the same hotkey** to keep the score rather than randomizing the selection. This approach was intentionally lenient.

Without this rule, duplicate submissions could have resulted in **all participating hotkeys receiving negative outcomes**, leaving the miner with no rewards at all.

Instead, the system allowed **one hotkey to retain valid rewards**, while penalizing the duplicates.

Policy Going Forward

This leniency should not be assumed in future cycles.

The current approach was chosen to avoid fully eliminating rewards for miners who had already contributed valid data under at least one hotkey. However, future duplication handling may adopt **randomized or stricter enforcement**, which could result in **all duplicated hotkeys receiving negative scores or no rewards**.

For this reason, miners are **strongly encouraged to operate a single hotkey when generating Face Variations**.

Submitting the same model outputs across multiple hotkeys will be treated as duplication and may lead to **watchlisting, reputation penalties, and the loss of emissions**.

Summary

One miner producing **high-quality, unique data** will earn more rewards over time.

Multiple miners running the **same model and submitting duplicated outputs** will trigger duplication detection, leading to **penalties, watchlisting, and loss of rewards**.

The incentive structure is therefore designed to reward **originality and sustained high-quality contributions**, not horizontal scaling of identical outputs.

Participation Policy (Sandbox vs Execution)

Phase 4 — Cycle 2 Sandbox (current stage)

- During the Sandbox stage, Face/UAV reputation is not used for emissions yet. Validators continue operating normally, but Face/UAV-based reward allocation is not activated in this stage.
- Sandbox is for calibration and stability checks before Face/UAV rewards go live.

Phase 4 — Cycle 2 Execution (UAV rewards go live)

- Once Execution begins, validators will start allocating the Face/UAV portion of rewards using the Phase 4 — Cycle 1 reputation results (each miner's `rep_score + tier`).
- At that point, miners with higher reputation tiers (e.g., Diamond) will receive stronger and longer-lasting UAV rewards because their `rep_score` starts higher and their tier multiplier is higher.

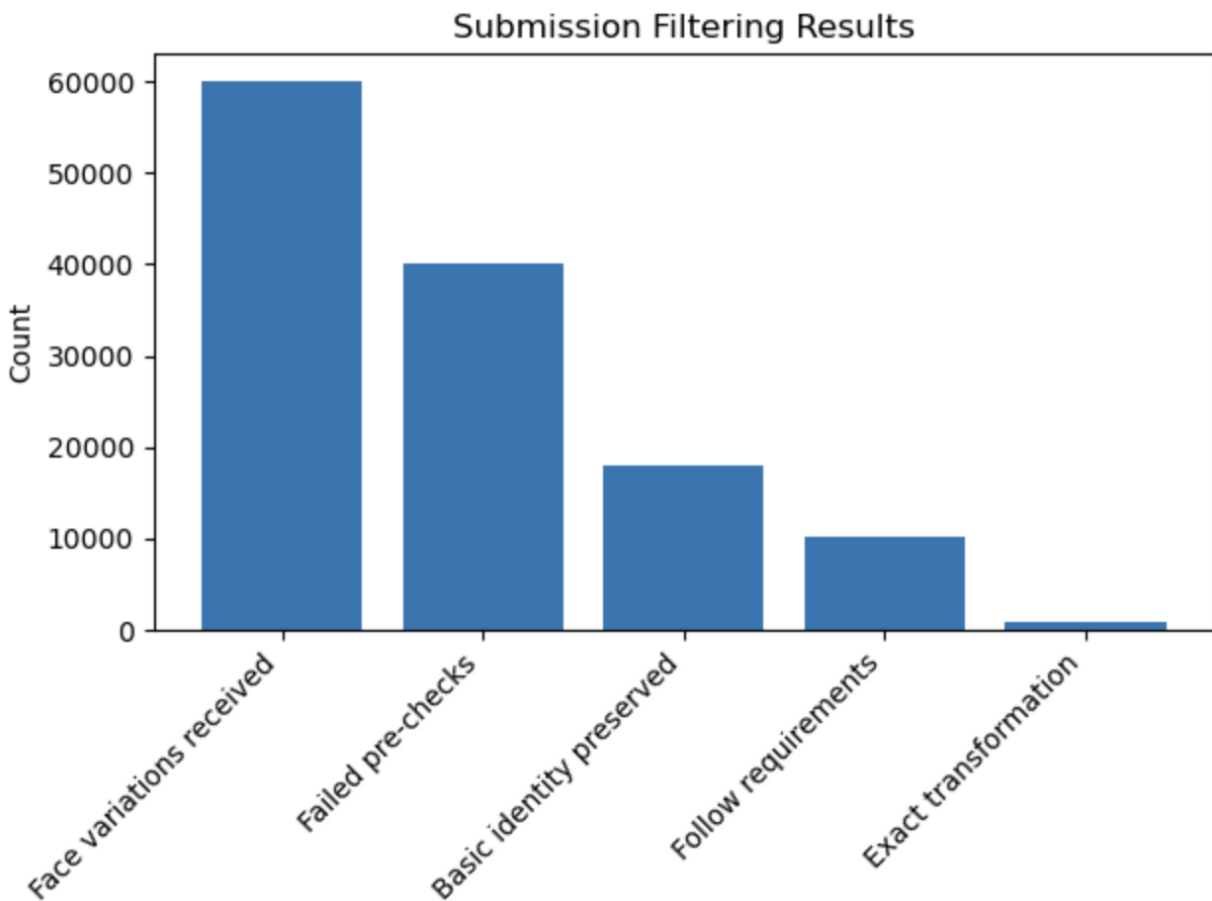
Reputation decay (applies once UAV rewards are live)

- Every time a miner receives a validator reward allocation, their `rep_score` will be reduced by 0.01 in the database.
- At the end of Phase 4 — Cycle 2 Execution, any remaining `rep_score` (if > 0) will be carried forward and combined with the next cycle's reputation update. Face/UAV-based rewards will resume in Phase 4 — Cycle 3 Execution, using the updated reputation snapshot.
- This means Watch-tier miners will decay to 0 faster and stop receiving Face/UAV rewards sooner, while higher-tier miners will remain eligible longer—unless they keep contributing validated Face/UAVs to rebuild score.

Tier & score rules

- A miner's **tier is fixed within a cycle** and only updates when the next UAV cycle snapshot is computed.
- `rep_score` is dynamic and will **trend downward through decay** as rewards are paid out.

Statistics on the Face variations



Face variations are currently one of the highest-load pipelines in the system. Their purpose is to generate constrained biometric transformations such as pose, lighting, expression, or background edits while preserving the identity of the validator-provided subject.

Like UAVs, this workflow remains heavily manual, and the volume illustrates why strong filtering and reputation mechanisms are essential.

Over **60,000 Face variations** were received.

More than **40,000** entries failed automatic pre-checks due to:

- corrupted or unreadable images

- duplicate of the original seed image
- duplicate of another miner's submission

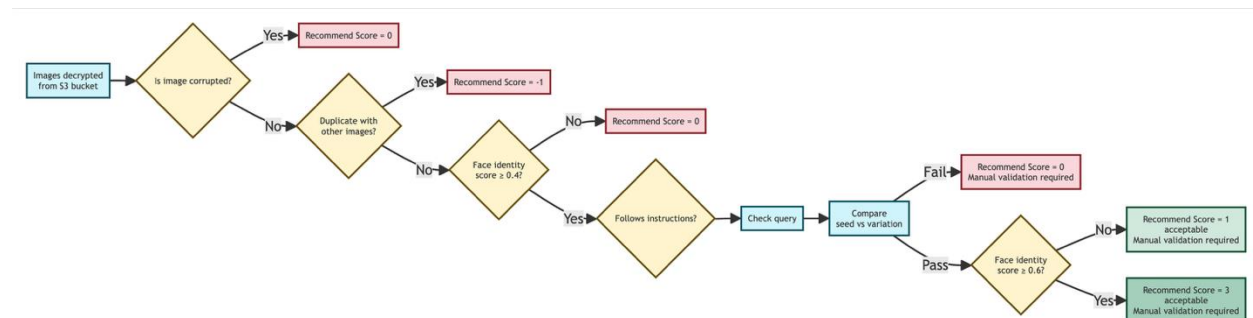
More than **18076** preserve identity at a basic recognizable level

Nearly **10267** follow the query requirements *and* preserve identity

More than **947** perform exactly the requested transformation, no more and no less, while preserving the identity of the seed image.

Flow chart on how Face Validation are Processed

Below is a small window into how manual validation is performed for face variation submissions.



Step 1: Filter out all the corrupt and duplicate images

Step 2: Identity is relatively preserved (AdaFace)

Step 3: Run the image to detect the variation type

Step 4: Sent to the manual validator checks for:

- Is the person still the same individual?
- Did the miner follow the query?
- Did the miner exploit?
- Does the image have more than variations?

- Are artifacts, overlays, or shortcuts present?

query_text:

For the face image provided, generate the following variations while preserving identity: 1. expression_edit (medium): Neutral to smile, serious, or mildly surprised expression 2. pose_edit (light): ±15° rotation (slight head tilt or turn) IMPORTANT: The subject's face must remain recognizable across all variations. Each variation should clearly address the specified type and intensity level.

POSE EDIT



BASE ID

REF-001

HOTKEY

EXECUTION ID

challenge_1770757392_

CHANGE HEAD POSE (YAW/PITCH/ROLL) WHILE KEEPING IDENTITY

±15° rotation (slight head tilt or turn)

SCORE (-5 - 5)

0

Score hint: 2

COMMENT

Partial match: More than one variation detected but there

EXPRESSSION EDIT



CHANGE FACIAL EXPRESSION WHILE PRESERVING IDENTITY

Neutral to smile, serious, or mildly surprised expression

SCORE (-5 - 5)

0

Score hint: 0

COMMENT

Label mismatch: No variation detected

challenge_1770757392_

TEMPORAL STAMP

2026-02-10 21:46:30+00:00

Partial match: More than one variation detected but there was a partial match

INTENSITY

light

IDENTITY PRESERVATION RATE

0.8759

Label_detected

type: multiple_variations

intensity:

all_detected: pose_edit medium expression_edit medium

Sent_deep_fake

is_deep_fake: false

deep_fake_score: N/A

Face_orientation_change

Yaw: 33.33567565242391

Roll: -169.9301996672336

Pitch: 5.097356223689766

final_response: Face in second photo has turned 31.0° to the RIGHT, and tilted head DOWN by 12.0°.

Facial_expression_change

Eyes: left: 0.863 right: 0.921

Mouth: 0.005

Smile: 0.004

final_response: Expression changes: Expression changed from 'frowning' to 'neutral'.

Lighting_analysis_change

Contrast: 80.7151940918977

Brightness: 184.05286121368408

final_response: Lighting conditions are similar in both photos.

Background_analysis_change

Complexity: 0.10230954277541836

final_response: Both photos have white/plain backgrounds

White background: true

Identity_verification_change

Similarity Score: 0.8201117647058824

COMMENT

Label mismatch: No variation detected

INTENSITY

medium

IDENTITY PRESERVATION RATE

0.7942

Label_detected

type: no_variation

intensity:

Sent_deep_fake

is_deep_fake: false

deep_fake_score: N/A

Face_orientation_change

Yaw: -0.6653997624190641

Roll: -179.26407757853846

Pitch: 17.339601271516454

final_response: Face orientation is similar in both photos (changes < 10°).

Facial_expression_change

Eyes: left: 0.724 right: 0.795

Mouth: 0.013

Smile: 0.012

final_response: Expression similar in both photos (both: frowning).

Lighting_analysis_change

Contrast: 80.08766372302547

Brightness: 182.31877040863037

final_response: Lighting conditions are similar in both photos.

Background_analysis_change

Complexity: 0.1182865692584251

final_response: Both photos have white/plain backgrounds

White background: true

Identity_verification_change

Similarity Score: 0.856589705882353

(Here is a sneak peak on what the manual validator sees)

Examples Of Face Variations Processed:

Rejected Identity not Perserved:

1. Seed Image
2. Expression Edit (Face identity below threshold ($\text{ada_sim}=0.322 < 0.4$))
3. Background Edit (Face identity below threshold ($\text{ada_sim}=0.313 < 0.4$))
4. Lighting Edit (Face identity below threshold ($\text{ada_sim}=0.308 < 0.4$))



1. Base Image
2. Expression Edit (Face identity below threshold ($\text{ada_sim}=0.33 < 0.4$))



Duplicates that will be rejected starting from Feb 16th :

If a miner just **copies and pastes** the **face** on a **different background**, they will start to get negative scores.

Examples:

1. Seed Image
2. Background edit (copy and paste of the face on different background)



1. Seed Image
2. Background edit (copy and paste of the face on different background)



Positive cases that still need Manual Review:

1. Seed Image
2. Pose edit (Partial match: More than one variation detected but there was a partial match, detected: pose edit – far and expression edit – medium, score hint = 2)

- 3. Expression edit (Partial match: More than one variation detected but there was a partial match, detected: pose edit - medium and expression edit – far, score hint = 2)
- 3. Lighting edit (Acceptable (ada_sim=0.822, mediapipe_sim=0.626); Label close: claimed 'lighting_edit' 'far', detected 'medium', score hint = 3)

